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Digital Systems -Tutorial 2

1. Given a pair of terms for which a variable appears in one term and the complement of that variable in another, the *consensus* term is formed by multiplying the two original terms together, leaving out the selected variable and its complement. For example, the consensus of *ab* and *a'c* is *bc*. We can show that the consensus term is redundant in a Boolean expression. That is, a Boolean expression of the form XY + X'Z + YZ is equivalent to XY + X'Z. Show this via a truth table.

2. Show that the following equation (in Boolean algebra) is valid.

$$A'BC'D+(A'+BC)(A+C'D')+BC'D+A'BC' = ABCD+A'C'D'+ABD+ABCD'+BC'D$$

3. Simplify $(A'B(D'+C'D)+B(A+A'CD))$ to one literal (a literal is a variable or its complement).

4. Simplify the Boolean expression $(A'+C)(A'+C')(A+B+C'D)$.

5. Given the Boolean function $F = xy'z + x'y'z + w'xy + wx'y + wxy$: (a) Obtain the truth table of the function. (b) Draw the logic diagram for the Boolean expression. (c) Simplify the function using Boolean Algebra. (d) Draw the logic diagram for the simplified form. and (e) Compare the total number of gates (inverters, OR gates and AND gates) for the two cases: the original Boolean function and the simplified form.

6. Find the complement of the following expressions: (a) $xy' + x'y$ and (b) $(AB' + C)D' + E$.

7. Consider the following Boolean function expressed as the sum of minterms.

$$F = A'BC + AB'C + AC'B'$$

Express this Boolean function as the product of maxterms.

8. Obtain the minterm expansion of the following Boolean function using a truth table. Write the solution using the $\sum$ notation. Also obtain the maxterm expansion and write the same using the $\prod$ notation.

$$F = X'(Y + Z') + XY'$$

9. Prove DeMorgan's laws (theorems) using a truth table.

10. The output of a majority circuit is 1 if a majority (more than half) of its inputs are equal to 1. The output is 0 otherwise. Construct a truth table for a three-input majority circuit and derive a simplified sum-of-products expression for its output.

1. Given a pair of terms for which a variable appears in one term and the complement of that variable in another, the *consensus* term is formed by multiplying the two original terms together, leaving out the selected variable and its complement. For example, the consensus of *ab* and *a'c* is *bc*. We can show that the consensus term is redundant in a Boolean expression. That is, a Boolean expression of the form XY + X'Z + YZ is equivalent to XY + X'Z. Show this via a truth table.

X Y Z	XY	YZ	X'Z	XY+X'Z+YZ	XY+X'Z
0 0 0	0	0	0	0	0
0 0 1	0	0	1	1	1
0 1 0	0	0	0	0	0
0 1 1	0	1	1	1	1
1 0 0	0	0	0	0	0
1 0 1	0	0	0	0	0
1 1 0	1	0	0	1	1
1 1 1	1	1	0	1	1

We can see that both the colums are matching hence they're equivalent.

2. Show that the following equation (in Boolean algebra) is valid.

$$A'BC'D+(A'+BC)(A+C'D')+BC'D+A'BC' = ABCD+A'C'D'+ABD+ABCD'+BC'D$$

LHS:

$A'BC'D + \cancel{AA'}^{0} + A'C'D' + ABC + \cancel{BCC'}^{0}D + BC'D + A'BC'$

$= A'C'D' + ABC + BC'D + A'C'(BD+B)$

$= A'C'D' + ABC + BC'D + A'BC'(D + \bar{D})$

$= A'C'D' + ABC + BC'D + A'BC'D + A'BC'D'$

$= A'C'D' + ABC + BC'D$

RHS:

$ABCD + A'C'D' + ABD + ABCD' + BC'D$

$= ABC(D + D') + A'C'D' + BC'D + ABD$

$= ABC + A'C'D' + BC'D + ABD\,(C + C')$

$= ABC + A'C'D' + BC'D + ABCD + ABDC'$

$= ABC + A'C'D' + BC'D + ABC'D$

$= ABC + A'C'D' + BC'D$

Hence LHS = RHS

3. Simplify $(A'B(D' + C'D) + B(A + A'CD))$ to one literal (a literal is a variable or its complement).

$A'BD' + A'BC'D + AB + A'BCD$

$= A'BD' + AB + A'BD$

$= A'B + AB$

$= B$

4. Simplify the Boolean expression $(A' + C)(A' + C')(A + B + C'D)$.

$(A'A' + A'C' + A'C + CC')(A + B + C'D)$

$= (A' + A')(A + B + C'D)$

$= A'(B + C'D)$

5. Given the Boolean function $F = xy'z + x'y'z + w'xy + wx'y + wxy$: (a) Obtain the truth table of the function. (b) Draw the logic diagram for the Boolean expression. (c) Simplify the function using Boolean Algebra. (d) Draw the logic diagram for the simplified form. and (e) Compare the total number of gates (inverters, OR gates and AND gates) for the two cases: the original Boolean function and the simplified form.

(a)

x	y	z	w	xy'z	x'y'z	w'xy	wx'y	wxy	F
0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	0	0
0	0	1	0	0	1	0	0	0	1
0	0	1	1	0	1	0	0	0	1
0	1	0	0	0	0	0	0	0	0
0	1	0	1	0	0	0	1	0	1
0	1	1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	1	0	1
1	0	0	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0	0	0
1	0	1	0	1	0	0	0	0	1
1	0	1	1	1	0	0	0	0	1
1	1	0	0	0	0	1	0	0	1
1	1	0	1	0	0	0	0	1	1

1 1 1 0 0 0 0 0 0 1

1 1 1 1 0 0 0 0 1 1

(c) $xy'z + x'y'z + w'xy + wx'y + wxy$

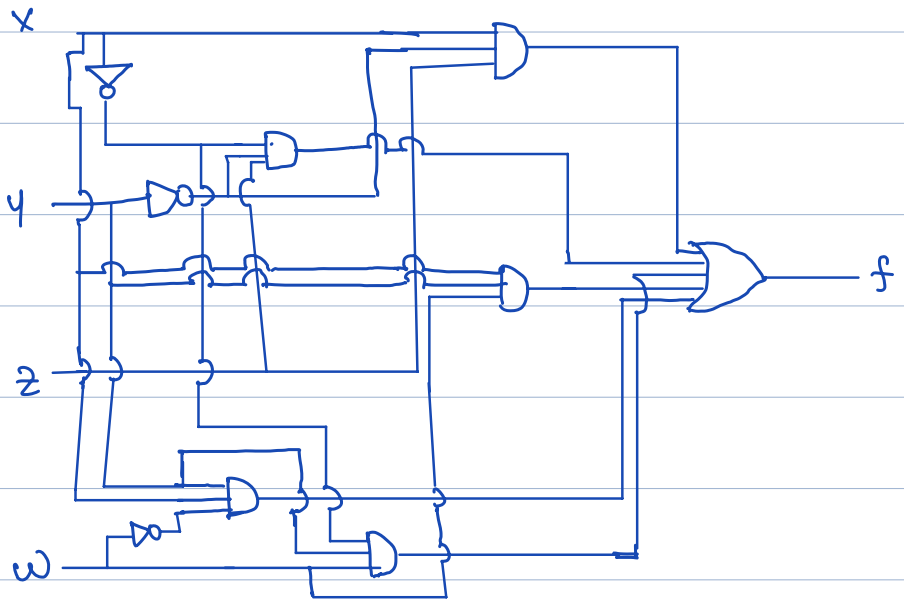
$= y'z + w'xy + wy(x+\bar{x})$

$= y'z + xy + wyx'$

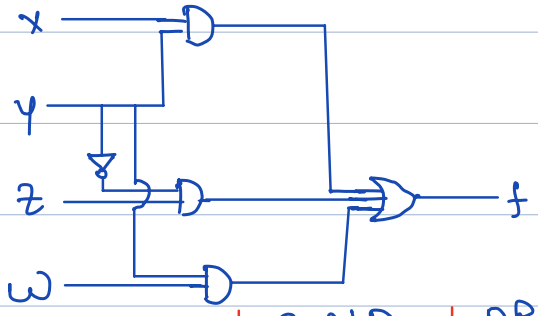
$= y'z + y(wx' + x(w+w') + xw)$

$= y'z + yw + yx$

(b)



(d)



(e)

	AND	OR	NOT
original	5	1	3
simplified	3	1	1

6. Find the complement of the following expressions: (a) $xy' + x'y$ and (b) $(AB' + C)D' + E$.

(a) $\overline{xy' + x'y} = \overline{xy'} \cdot \overline{x'y} = (x'+y)(x+y')$

$= xy + x'y'$

(b) $\overline{(AB'+C)D'+E} = \overline{(AB'+C)D'} \cdot E' = (\overline{AB'+C} + D)E'$

$= (\overline{AB'} \cdot \overline{C} + D)E' = ((A'+B)C' + D)E'$

7. Consider the following Boolean function expressed as the sum of minterms.

$$F = A'BC + AB'C + AC'B'$$

Express this Boolean function as the product of maxterms.

$A'BC + AB'C + AC'B'$

$(A+B+C)(A+\bar{B}+C)(A+B+\bar{C})(\bar{A}+\bar{B}+C)(\bar{A}+\bar{B}+\bar{C})$

8. Obtain the minterm expansion of the following Boolean function using a truth table. Write the solution using the $\sum$ notation. Also obtain the maxterm expansion and write the same using the $\prod$ notation.

$$F = X'(Y + Z') + XY'$$

$F = x'(y+z') + xy'$

$x'y + x'z' + xy'$

$= x'y(z+z') + x'z'(y+y') + xy'(z+z')$

$\Sigma = x'yz + x'yz' + x'z'y' + xy'z + xy'z'$

$\Pi: (x'+y'+z')(x+y+z')(x'+y'+z)$

9. Prove DeMorgan's laws (theorems) using a truth table.

(a) $\overline{xy} = \bar{x} + \bar{y}$

x	y	$\overline{xy}$	$\bar{x}+\bar{y}$	$\bar{x}\bar{y}$	$\overline{x+y}$

(b) $\overline{X+Y} = \overline{X}\,\overline{Y}$

0	0	1	1	1	1
0	1	1	1	0	0
1	0	1	1	0	0
1	1	0	0	0	0

10. The output of a majority circuit is 1 if a majority (more than half) of its inputs are equal to 1. The output is 0 otherwise. Construct a truth table for a three-input majority circuit and derive a simplified sum-of-products expression for its output.

X	Y	Z	M(X,Y,Z)
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$M(X,Y,Z) = \overline{X}YZ + X\overline{Y}Z + XY\overline{Z} + XYZ$

$= XY + YZ + ZX$